$$\nabla \times \vec{E} = -\frac{\partial}{\partial t}\vec{B}$$

$$\vec{E} = \vec{E}_0 e^{i(\vec{k}_1 \cdot \vec{r} - \omega_1 t)}$$

$$\vec{B} = \vec{B}_0 e^{i(\vec{k}_2 \cdot \vec{r} - \omega_2 t)}$$

$$\nabla \times = ik\times$$

$$\frac{\partial}{\partial t} = -i\omega$$

$$\nabla \times A = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ A_x & A_y & A_z \end{vmatrix} \qquad \vec{A} = (A_1, A_2, A_3) e^{i(k_1 x + k_2 y + k_3 z - \omega t)}$$

$$= \left(\frac{\partial A_3}{\partial y} - \frac{\partial A_2}{\partial z}\right)\vec{i} + \left(\frac{\partial A_1}{\partial z} - \frac{\partial A_3}{\partial x}\right)\vec{j} + \left(\frac{\partial A_2}{\partial x} - \frac{\partial A_1}{\partial y}\right)\vec{k}$$

$$= (ik_2 A_3 - ik_3 A_2)\vec{i} + (ik_3 A_1 - ik_1 A_3)\vec{j} + (ik_1 A_2 - ik_2 A_1)\vec{k}$$

$$= i \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ k_1 & k_2 & k_3 \\ A_1 & A_2 & A_3 \end{vmatrix} = i\vec{k} \times \vec{A}$$

Maxwel eq: $\nabla \times \vec{E} = -\frac{\partial}{\partial t}\vec{B}$

$$i\vec{k} \times \vec{E} = i\omega \vec{B}$$

$B \perp \vec{k}、\vec{E}$ 组成平面.

泊松方程：

$$\nabla\cdot\vec{E}=\frac{\rho}{\varepsilon_0}\qquad \nabla\times\vec{E}=-\frac{\partial}{\partial t}\vec{B}_0=0$$

注意到数学上：对标量 φ.　　$\nabla\times(\nabla\varphi)=0$

∴我们构造　　$E=-\nabla\varphi$

令　　$\varphi(\infty)=0$.

由 Maxwel eqs.　　$\nabla\cdot\vec{E}=\rho/\varepsilon_0 \Longrightarrow \nabla\cdot(\nabla\varphi)=\nabla^2\varphi=-\rho/\varepsilon_0$

∴

$$\nabla^2\varphi_1=+\frac{en_{e_1}}{\varepsilon_0}$$

$$\begin{aligned}\vec{E}\cdot d\vec{l} &= -\left(\frac{\partial\varphi}{\partial x}\hat{x}+\frac{\partial\varphi}{\partial y}\hat{y}+\frac{\partial\varphi}{\partial z}\hat{z}\right)\cdot(\hat{x}dx+\hat{y}dy+\hat{z}dz)\\ &= -\left(\frac{\partial\varphi}{\partial x}dx+\frac{\partial\varphi}{\partial y}dy+\frac{\partial\varphi}{\partial z}dz\right).\\ &= -d\varphi\end{aligned}$$

对任2点，P_1、P_2：

$$\int_{P_1}^{P_2}\vec{E}\cdot d\vec{l}=\varphi_2-\varphi_1$$　与路径无关.

cold ion

$n = n_i + n_e$, $n_i = n_0$. $n_e = n_{e0} + n_{e1}$ $\rho = -n_{e1} e$

$v_e = \vec{v}_0 + \vec{v}_1$, $\vec{E} = \vec{E}_0 + \vec{E}_1$

$\vec{B} = \vec{B}_0 + \vec{B}_1 + \cdots$ $\begin{cases} \vec{B}_1 = 0 : \text{静电波.} \\ \text{else} : \text{一般电磁波.} \end{cases}$

扰动磁场.

$\vec{k}$ 与 $\vec{B}_0$: $\vec{k} \parallel \vec{B}_0$: 平行. $\vec{k} \perp \vec{B}_0$: 垂直.

$\vec{k}$ 与 $\vec{E}$: $\vec{k} \parallel \vec{E}$: 纵波. $\vec{k} \perp \vec{E}$: 横波.

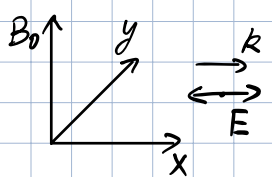
考虑 Maxwel eq : $\nabla \times \vec{E}_1 = -\frac{\partial}{\partial t} \vec{B}_1$

$\nabla \times \longrightarrow i\vec{k} \times$, $\frac{\partial}{\partial t} \longrightarrow -i\omega$.

$\vec{k} \times \vec{E}_1 = \omega \vec{B}_1$

- 纵 : $\vec{k} \times \vec{E}_1 = 0$. $\vec{B}_1 = 0$ 静电波.
- 横 : 电磁波.

垂直于 B_0 的静电电子振荡. $\vec{k} \perp \vec{B}_0$, $\vec{k} \parallel \vec{E}_1$. 纵波. $\vec{B}_1 = 0$



完整流体方程：

离子+电子：　$\rho = n_i q_i + n_e q_e$

cold ion assumption：$n_i = n_0$.

$\rho = q_i n_i - q_e n_e = (n_0 - n_e)e$

离子很重，不以所包含频率运动，形成一固定均匀正电荷本底.

$\vec{j} = n_i q_i \vec{v}_i + n_e q_e \vec{v}_e$

Maxwel eqs：　$\nabla \cdot \vec{E} = \frac{\rho}{\varepsilon_0} = \frac{(n_i - n_e)e}{\varepsilon_0} = -\frac{n_{e1} e}{\varepsilon_0}$

$\nabla \times \vec{E} = -\frac{\partial}{\partial t}\vec{B}$　✗ 纵波.

$\nabla \cdot \vec{B} = 0$.　✗ $\vec{B}_1 = 0$

$\nabla \times \vec{B} = \mu_0 (n_i q_i \vec{v}_i + n_e q_e \vec{v}_e) + \frac{1}{c^2}\frac{\partial}{\partial t}\vec{E}$.　✗ $\vec{B}_1 = 0$

流体动力学方程：　无热运动，$KT = 0$.

$m_j n_j [\frac{\partial \vec{v}_j}{\partial t} + (\vec{v}_j \cdot \nabla)\vec{v}_j] = q_j n_j (\vec{E} + \vec{v}_j \times \vec{B}) - \nabla \cdot \vec{P}_j + f_{\text{碰撞}}$　$j = i, e$.

高次项=0　　压力 $P = nKT$

连续性方程：

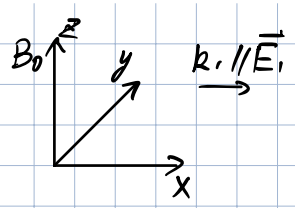
$\frac{\partial n_j}{\partial t} + \nabla \cdot (n_j \vec{v}_j) = 0$.　$j = i, e$.

仅剩 $n_0 \nabla \cdot \vec{v}_{e1}$

状态方程：　绝热：$\gamma = (2+N)/N$. 一维 $\gamma = 3$.

$P = C (m_j n_j)^{\gamma_j}$，　$j = i, e$.

$\gamma_i = \frac{C_p}{C_v}$　无热运动.



最后：

$$\begin{cases} m\dfrac{\partial \vec{v}_{e1}}{\partial t} = -e(\vec{E}_1 + \vec{v}_{e1}\times\vec{B}_0) & ① \\ \dfrac{\partial n_{e1}}{\partial t} + n_0 \nabla\cdot\vec{v}_{e1} = 0 & ② \\ \nabla\cdot\vec{E}_1 = \dfrac{e(n_i - n_e)}{\varepsilon_0} = -\dfrac{e n_{e1}}{\varepsilon_0} & ③ \end{cases}$$

① 展成分量：

$$\begin{cases} m\dfrac{\partial v_{ex1}}{\partial t} = -eE_1 - v_{ey1}B_0 = -i\omega m v_{ex1} \\ m\dfrac{\partial v_{ey1}}{\partial t} = +e v_{ex1}B_0 = -i\omega m v_{ey1} \\ m\dfrac{\partial v_{ez1}}{\partial t} = 0 = -i\omega m v_{ez1} \end{cases}$$

$$v_{ey1} = \frac{ieB_0}{\omega m} v_{ex1}$$

$$-\frac{ie^2B_0^2}{\omega m} v_{ex1} - eE_1 = -i\omega m v_{ex1}$$

$$\Leftrightarrow \quad i\left(\omega m - \frac{e^2B_0^2}{\omega m}\right) v_{ex1} = eE_1$$

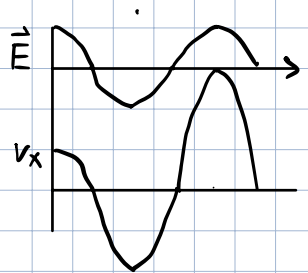
$$\therefore \quad v_{ex1} = \frac{eE_1}{(\omega m - \frac{e^2B_0^2}{\omega m})i} = \frac{eE_1/i\omega m}{[1-(\frac{eB_0}{m}/\omega)^2]}$$

$$= \frac{eE_1/i\omega m}{1-(\omega_c/\omega)^2}$$

$\omega_c = \dfrac{eB_0}{m}$ $T_c = \dfrac{2\pi}{\omega_c}$

拉莫回旋

回旋共振：$\omega = \omega_c$ $v_x \longrightarrow \infty$.



$\vec{E}_1$ 与 v_{x1} 同时变号使电子不停加速.

$$v_{ey1} = \frac{i\overbrace{eB_0}^{\omega_c}}{\omega m} v_{ex1} = i\frac{\omega_c}{\omega} v_{ex1} = \frac{eE_1/\omega m}{1-(\omega_c^2/\omega^2)} \cdot \frac{\omega_c}{\omega}$$

②: $\frac{\partial n_{e1}}{\partial t} + n_0 \nabla \cdot \vec{v}_{e1} = 0$. $\quad -i\omega n_{e1} + n_0 (\underbrace{i\vec{k}}\cdot \vec{v}_{e1}) = 0$

$\vec{k}$ 沿 x 轴

$$n_{e1} = \frac{k}{\omega} n_0 v_{ex1}$$

③: $\nabla \cdot \vec{E}_1 = \frac{e}{\varepsilon_0}(n_i - n_e) = -\frac{e n_{e1}}{\varepsilon_0}$

$$ikE_1 = -\frac{e n_{e1}}{\varepsilon_0} = -\frac{e}{\varepsilon_0}\frac{k}{\omega} n_0 \underbrace{v_{ex1}}$$

$$= -\frac{e}{\varepsilon_0}\frac{k}{\omega} n_0 \frac{eE_1/i\omega m}{(1-\omega_c^2/\omega^2)}$$

$$\Leftrightarrow \quad 1 = \frac{e^2}{\varepsilon_0}\frac{n_0}{\omega^2 m} \cdot \left(1-\frac{\omega_c^2}{\omega^2}\right)^{-1}.$$

$$\omega^2 - \omega_c^2 = \underbrace{\frac{e^2 n_0}{\varepsilon_0 m}} \equiv \omega_p^2.$$

色散关系有: $\omega^2 = \omega_c^2 + \omega_p^2 \equiv \omega_h^2$ 上杂化频率.

等离子体振荡频率 ω_p^2 + 拉莫回旋频率 ω_c^2.

- 穿过 $\vec{B}$ 的静电电子波有 ω_h.
- 沿 $\vec{B}$ 传播, 只有 ω_p. 无 ω_c.

轨迹？ $v_x = \frac{1}{1-\omega_c^2/\omega^2} \frac{e\tilde{E}_1}{im\omega}$

$e^{i(kx-\omega t)}$

$\cos(kx-\omega t) + i\sin(kx-\omega t)$

$= \frac{1}{1-\omega_c^2/\omega^2} \frac{e}{m\omega} |E_1| \sin(kx-\omega t)$

$v_y = \frac{\omega_c}{\omega} \frac{1}{1-\omega_c^2/\omega^2} \frac{e}{m\omega} |E_1| \cos(kx-\omega t)$.

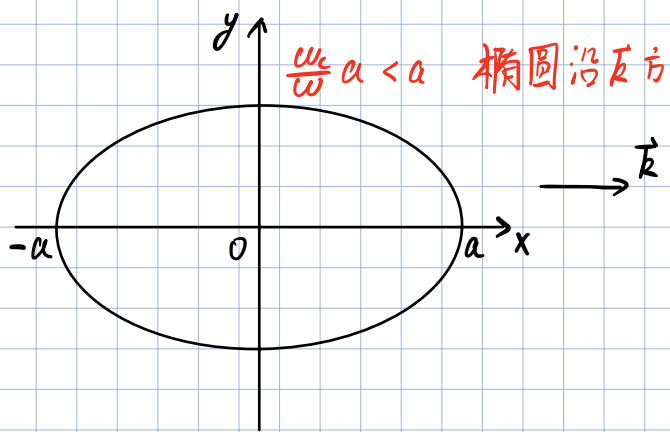
令 $x = x_0$,

$x_1 = \frac{1}{1-\omega_c^2/\omega^2} \frac{e|E_1|}{m\omega} \cos(kx_0 - \omega t)$.

$y_1 = -\frac{\omega_c}{\omega} \frac{1}{1-\omega_c^2/\omega^2} \frac{e|E_1|}{m\omega} \sin(kx_0 - \omega t)$

轨迹. $x_1^2 + \frac{y_1^2}{(\frac{\omega_c}{\omega})^2} = \left(\frac{1}{1-\omega_c^2/\omega^2} \frac{e|E_1|}{m\omega}\right)^2$ 椭圆.

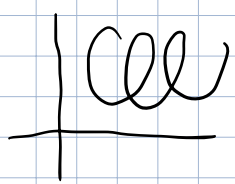
a



$\frac{\omega_c}{\omega} a < a$ 椭圆沿 x 方向被拉长.

说明：

- 无热运动，轨迹闭合，群速度 $v_g = 0$.
  类似于电子静电振荡.
- 考虑电子温度. 求解困难. 可模拟，曲线不闭合.



Q: 上杂化频率出现的原因?

恢复力 = $-q(\vec{E}+\vec{v}\times\vec{B})$. 回复力↗使振荡频率↗.

Case1 不是等离子体，单个电子. 简单的 Lorentz 回旋.

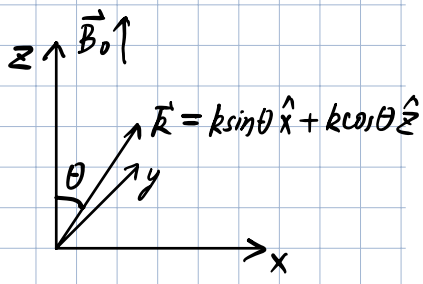
$n_0 \to 0$ ω_c 不变. $\omega_p = \left(\frac{e^2 n_0}{\varepsilon_0 m}\right)^{\frac{1}{2}} \to 0$.

$\omega_h \longrightarrow \omega_c$

Case 2. 无外磁场. 简单的静电振荡.

$B_0 \to 0$ $\omega_c \longrightarrow 0$ ω_p 不变.

$\vec{k} \perp \vec{B}_0 \longrightarrow \vec{k}$ 与 B_0 夹角为 θ



纵波. $\vec{E}_1 = E_1\sin\theta\,\hat{x} + E_1\cos\theta\,\hat{z}$

$$\begin{cases} m\dfrac{\partial \vec{v}_{e1}}{\partial t} = -e(\vec{E}_1 + \vec{v}_{e1}\times\vec{B}_0) & ① \\ \dfrac{\partial n_{e1}}{\partial t} + n_0\nabla\cdot\vec{v}_{e1} = 0 & ② \\ \nabla\cdot\vec{E}_1 = \dfrac{e(n_i - n_e)}{\varepsilon_0} = -\dfrac{e n_{e1}}{\varepsilon_0} \Rightarrow \nabla^2\varphi = +\dfrac{e n_{e1}}{\varepsilon_0} & ③ \end{cases}$$

$\Rightarrow m\dfrac{\partial \vec{v}_{e1}}{\partial t} = -e(-\nabla\varphi + \vec{v}_{e1}\times\vec{B}_0)$

$m\dfrac{\partial \vec{v}_{e1}}{\partial t} = e(\vec{E}_1 - \vec{v}_{e1}\times\vec{B}_0)$ 分量形式:

$$\begin{cases} m\dfrac{\partial v_{ex1}}{\partial t} = eE_1\sin\theta - e v_{ey1}B_0 \\ m\dfrac{\partial v_{ey1}}{\partial t} = e v_{ex1}B_0 \\ m\dfrac{\partial v_{ez1}}{\partial t} = eE_1\cos\theta \end{cases}$$

$-i\omega$

$$v_{ex1} = \frac{eE_1\sin\theta}{(\omega m - \frac{e^2B_0^2}{\omega m})i} = \frac{eE_1\sin\theta / i\omega m}{[1 - (\frac{eB_0}{m}/\omega)^2]}$$

拉莫回旋 $\omega_c = \dfrac{eB_0}{m}$, $T_c = \dfrac{2\pi}{\omega_c}$

$$= \frac{eE_1\sin\theta/i\omega m}{1 - (\omega_c/\omega)^2},$$

$$v_{ey1} = \frac{ieB_0}{\omega m}v_{ex1} = i\frac{\omega_c}{\omega}v_{ex1} = \frac{eE_1\sin\theta/\omega m}{1-(\omega_c^2/\omega^2)}\cdot\frac{\omega_c}{\omega}$$

$$v_{ez1} = -\frac{ieE_1\cos\theta}{\omega m}$$

② : $\frac{\partial n_{e1}}{\partial t} + n_0 \nabla \cdot \vec{v}_{e1} = 0$ $\quad -i\omega n_{e1} + i n_0 \cdot (k_x v_{e1x} + k_z v_{e1z}) = 0$.

$$n_{e1} = \frac{n_0 k}{\omega} (v_{e1x} \sin\theta + v_{e1z} \cos\theta)$$

③ : $\nabla \cdot \vec{E}_1 = -\frac{e n_{e1}}{\varepsilon_0} = ik E_1 (\sin^2\theta + \cos^2\theta)$

$$= -\frac{e}{\varepsilon_0} \frac{n_0 k}{\omega} (v_{e1x} \sin\theta + v_{e1z} \cos\theta)$$

$$= -\frac{e}{\varepsilon_0} \frac{n_0 k}{\omega} \left(\frac{e E_1 \sin^2\theta / i\omega m}{1 - (\omega_c/\omega)^2} - \frac{ie E_1 \cos^2\theta}{\omega m} \right)$$

$$= -\frac{e^2 E_1 n_0 k}{\varepsilon_0 \omega^2 m} \left(\frac{\sin^2\theta}{i[1 - (\omega_c/\omega)^2]} - i\cos^2\theta \right)$$

$$(\sin^2\theta + \cos^2\theta) = \frac{e^2 n_0}{\varepsilon_0 \omega^2 m} \left(\frac{\sin^2\theta}{[1 - (\omega_c/\omega)^2} + \cos^2\theta \right).$$

$$= \frac{\omega_p^2}{\omega^2} \frac{\sin^2\theta + \cos^2\theta - (\frac{\omega_c}{\omega})^2 \cos^2\theta}{1 - (\frac{\omega_c}{\omega})^2}$$

$$\omega^2 (1 - \frac{\omega_c^2}{\omega^2}) = \omega_p^2 (1 - \frac{\omega_c^2}{\omega^2} \cos^2\theta)$$

$$(\omega^2 - \omega_c^2) = \omega_p^2 (1 - \frac{\omega_c^2}{\omega^2} \cos^2\theta)$$

$$\omega^2 - \omega_c^2 = \omega_p^2 - \frac{\omega_c^2 \omega_p^2}{\omega^2} \cos^2\theta$$

$$(\omega^2)^2 - \omega^2 \omega_h^2 + \omega_c^2 \omega_p^2 \cos^2\theta = 0$$

解出 ω :

$$\omega^2 = \frac{\omega_h^2 \pm \sqrt{\omega_h^4 - 4\omega_c^2 \omega_p^2 \cos^2\theta}}{2}$$

① when $\theta \to 0$. $\omega^2 = \frac{\omega_h^2 \pm \sqrt{\omega_h^4 - 4\omega_c^2 \omega_p^2}}{2} = \frac{\omega_h^2 \pm |\omega_p^2 - \omega_c^2|}{2}$

$\vec{B} // z$ 轴

$$= \frac{(\omega_p^2+\omega_c^2) \pm |\omega_p^2-\omega_c^2|}{2} \qquad \begin{cases} 1: + \\ 2: - \end{cases}$$

∴ i 若 $\omega_c \geq \omega_p$, $\omega_1 = \omega_c$, $\omega_2 = \omega_p$

ii 若 $\omega_c \leq \omega_p$, $\omega_1 = \omega_p$, $\omega_2 = \omega_c$

不可能有 $\cos\theta = 0$ (即 $\vec{B} \perp$) 情形下的 Lamour 回旋 ω_c 无意义.

∴ 做一个简单的等离子体振荡.

② when $\theta \to \frac{\pi}{2}$. $\omega^2 = \frac{\omega_h^2 \pm \omega_h^2}{2}$, $\omega_1^2 = \omega_h^2$. $\omega_2^2 = 0$.

上杂化等离子体振荡. $\omega_2 = 0$ 无意义.

Q: ω_1, ω_2 与 ω_c, ω_p, ω_h 大小关系?

$$\omega_1^2 = \frac{\omega_h^2 + \sqrt{\omega_h^4 - 4\omega_c^2\omega_p^2\cos^2\theta}}{2} \leq \omega_h^2, \quad \omega_1 \leq \omega_h.$$

i $\omega_c > \omega_p$. $\cos^2\theta \in [0,1]$.

$$\omega_1^2 = \frac{\omega_h^2 + \sqrt{\omega_h^4 - 4\omega_c^2\omega_p^2\cos^2\theta}}{2} \geq \frac{\omega_h^2 + \sqrt{\omega_h^4 - 4\omega_c^2\omega_p^2}}{2}$$

$$= \frac{\omega_c^2 + \omega_p^2 + |\omega_c^2 - \omega_p^2|}{2} = \omega_c^2. \quad (\omega_1 \geq \omega_c)$$

$$\omega_2^2 = \sim \leq \frac{(\omega_c^2 + \omega_p^2) - |\omega_c^2 - \omega_p^2|}{2} = \omega_p^2 \quad (\omega_2 \leq \omega_p).$$

ii $\omega_p > \omega_c$ 同理.

$$\omega_1 \geq \omega_p. \quad \omega_2 \leq \omega_c.$$

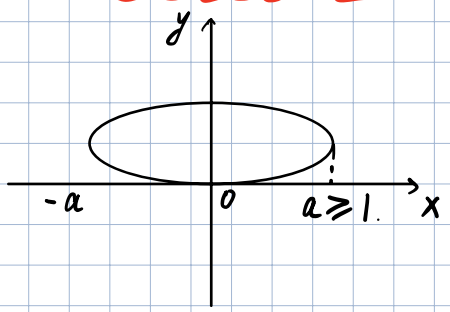
关系 $(\omega^2)^2 + \omega_c^2\omega_p^2\cos^2\theta = \omega^2\omega_h^2$ 为一个椭圆.

$$(\omega^2)^2 - \omega^2\omega_h^2 + \frac{1}{4}\omega_h^2 + \omega_c^2\omega_p^2\cos^2\theta = \frac{1}{4}\omega_h^2.$$

$$(\omega^2 - \frac{1}{2}\omega_h^2)^2 + \omega_c^2\omega_p^2\cos^2\theta = \frac{1}{4}\omega_h^4.$$

$$\frac{\cos^2\theta}{\left(\frac{\omega_h^2}{2\omega_c\omega_p}\right)^2} + \left(\frac{2\omega^2}{\omega_h^2} - 1\right)^2 = 1.$$

$$\frac{x^2}{a^2} + (y-1)^2 = 1$$



$$a = \frac{\omega_h^2}{2\omega_c\omega_p} = \frac{\omega_c^2 + \omega_p^2}{2\omega_c\omega_p}$$

$$= \frac{1}{2}\left(\frac{\omega_c}{\omega_p} + \frac{\omega_p}{\omega_c}\right) \geq 1.$$